GRIDLY: SEARCHING THE HISTORY OF FORMULA 1

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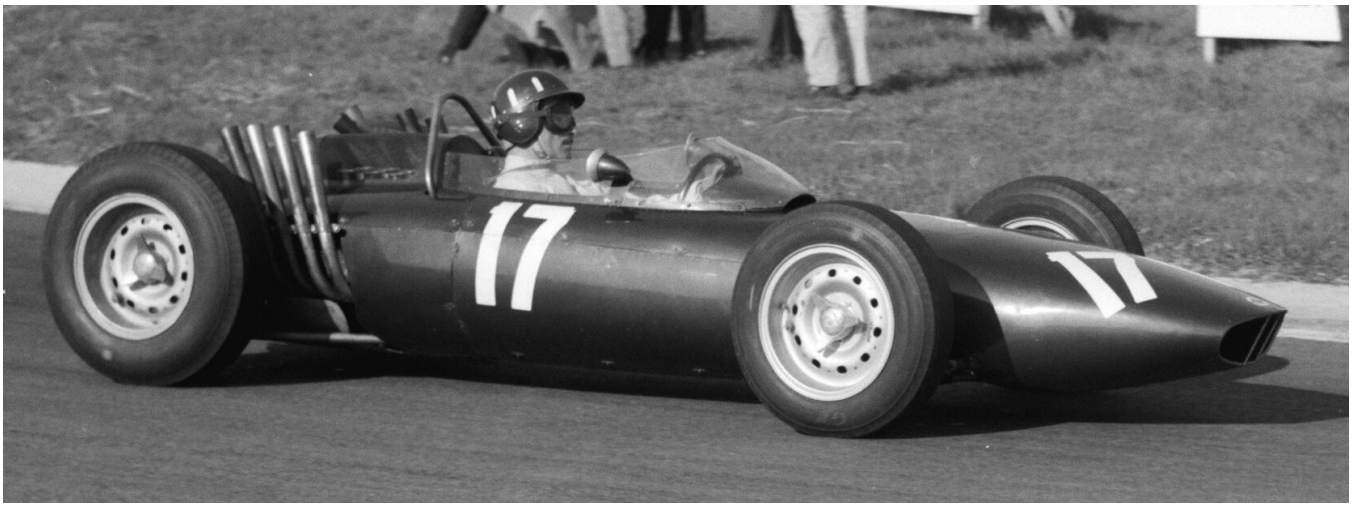


Figure 1: Graham Hill in the BRM P57 at the 1962 Dutch Grand Prix, Zandvoort, the race that gave him his first World Championship win. Photo by Joop van Bilsen, Anefo / Nationaal Archief (CC0) [10].

ABSTRACT

Formula One history is scattered across many sources. We build a search system that unifies it, combining race results and championship records from Jolpica F1 with English Wikipedia text, linked through Jolpica's article URLs. A reproducible pipeline collects, cleans and links the sources into 2058 documents (races, drivers, teams and circuits) from 1950 to 2026, totalling 2.8 million words.

CCS Concepts

• **Information systems** → Information retrieval query processing.

Keywords

Formula One, Motorsport, Information Retrieval, Search System, Wikipedia, Jolpica F1, Data Preparation, Data Quality, Conceptual Model, Collection Characterization

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1 INTRODUCTION

This work was developed for the course "Information Processing and Retrieval" (PRI) of the Master's in Informatics and Computing Engineering (M.EIC) at the Faculty of Engineering of the University of Porto (FEUP).

The choice of Formula One is motivated both by personal interest and by the nature of its data. Most of us follow the sport, but as relatively recent fans we wanted an easy way to learn about its history, which goes back to the first World Championship race in 1950. Information about its history comes in two forms: structured data, such as results and championship standings, and free text, such as race reports and driver biographies. Gaining deeper insight into a given topic often requires integrating both types of information, as they provide complementary perspectives. For example, finding races decided by a late mechanical failure requires matching the retirement reason in the results with the narrative of the race, which makes Formula One a good domain for an information retrieval system close to a real-world scenario.

Section 2 describes the dataset, including its sources, preparation, quality, structure, and characterization. Section 3 presents the search scenarios, and Section 4 concludes and outlines future work.

2 DATASET

2.1 Data Origin and Description

Two complementary sources were selected, which, when combined, cover the full history of the sport: the Jolpica F1 database and English Wikipedia.

Table 1 characterizes both sources in their raw form. For Jolpica, rows and columns are summed over its 11 tables; for Wikipedia, each row corresponds to an article, and each column to a field stored for it.

Source	Format	Files	Rows	Columns	Size (MB)
Jolpica F1	CSV	11	141029	110	7.30
Wikipedia	JSON	2225	2225	7	77.37

Table 1: Initial data sources characterization [3, 11]

Jolpica F1 [3] is an open-source project that succeeds the Ergast API, which was discontinued at the end of 2024. Its relational database holds the results of every World Championship race since the inaugural 1950 season, together with additional information such as starting grids, finishing positions, retirement reasons, team lineups, and championship standings. We used its CSV dump instead of the REST API, which would require around 260 paginated requests under an hourly rate limit. Wikipedia [11] provides the text: the narrative of each race, the biography of each driver and the history of each team and circuit. The articles were obtained in wikitext through the MediaWiki Action API [7], returning several articles per request.

The two sources are linked directly through Jolpica, which stores the Wikipedia URL of each race, driver, team and circuit. This avoids matching by name, which would be error-prone for drivers with the same name or races whose title changed over the years. Both sources are released under open licenses (CC BY-NC-SA 4.0 for Jolpica and CC BY-SA 4.0 for Wikipedia) that permit their use in an academic project.

2.2 Data Preparation

The pipeline (Figure 2) transforms the two raw sources into a validated collection of documents. It is implemented in Python as five scripts, each reading the output of the previous one. Raw downloads, intermediate files and the final collection are kept in separate folders.

Data Ingestion downloads the Jolpica dump, keeps the 11 tables needed and records the version used (12 September 2026), since the dump link always points to the latest one. *Entity Structuring* then joins these tables with Pandas [6] into one record per race, driver, team and circuit, with values such as the winner and retirements of a race or the career totals of a driver. Jolpica has one team row per chassis–engine combination ("Cooper-Climax", "Cooper-Maserati"), so rows that point to the same article were merged into one team, and only races already held were kept.

Each record carries the link to its Wikipedia article, which the *Wikipedia Crawler* uses to download the text through the MediaWiki API. Since Wikipedia blocks anonymous clients that send one request per page (HTTP 429), articles are requested in batches of 20, reducing about 2230 requests to 115. Each article is cached locally with its page and revision identifiers, which identify the exact version used and allow later runs to download only what is missing.

The *Wikitext Parser* converts the wikitext to clean text with mwparserfromhell [4], removing tables, references, infoboxes and images and keeping the visible text of links. In *Schema Mapping & Filtering*, each article is split by its sections, which are mapped by keywords to a fixed set of text fields per document type (Table 4), and sections without prose, such as classification tables, are discarded. Documents with fewer than 100 words of text are rejected, with the reason recorded. The *Relational Linker* links the accepted ones to each other in both directions (races, drivers and teams, and races and circuits), keeping only links between documents that exist in the collection, and exports them in JSON and CSV. Finally, *Collection Characterization* computes the statistics and plots of Section 2.6, using Matplotlib [2], NumPy [1] and scikit-learn [8].

The whole pipeline runs with a single `make` command, which creates the virtual environment, installs the dependencies and runs the five steps in order. Each step also has its own target (`collect`, `structure`, `wikipedia`, `documents` and `analyze`), while `make clean` removes the generated files but keeps the downloads, and `make clean-all` removes everything.

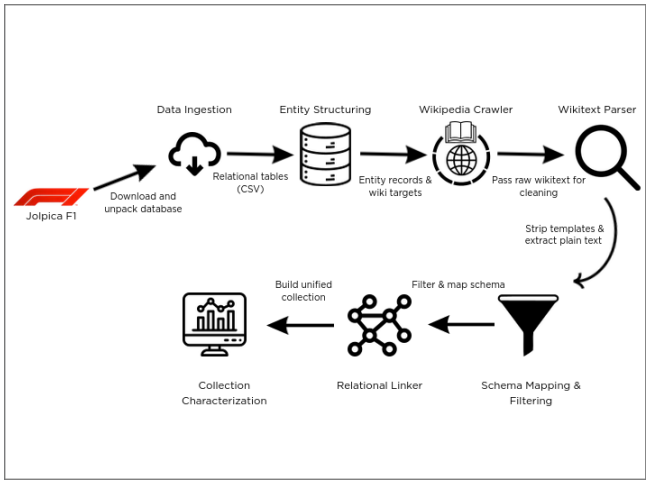


Figure 2: Data preparation pipeline

2.3 Data Quality

Table 2 lists the problems found and how they were handled. Most come from the links between the sources. In 28 cases, the Jolpica link leads to a **disambiguation page** ("Tony Brooks may refer to…"), usually because a more famous person has the same name. So that the fix survives a new run of the pipeline, they are resolved automatically by choosing the entry that mentions "Formula One" or "racing driver" (e.g., "Tony Brooks (racing driver)"), and each correction is recorded with the original link. The Long Beach circuit links to the article about the city, which is a valid article and cannot be detected automatically, so it was corrected by hand.

In the results, the **status** is a numeric code with no documentation. By grouping the values of the `detail` column ("Engine", "Spun

off", "+1 Lap") under each code, we inferred its meaning: finished, finished with laps behind, retired by accident, retired by failure, disqualified and did not start. This made it possible to keep the retirement reasons, one of the most useful structured fields.

Most of the 164 **short documents** are drivers from the 1950s with one or two races, whose article is a single sentence followed by a table. The threshold was chosen by inspecting these articles: raising it to 150 words would reject 111 more documents, many of them with useful text.

Issue	Count	Handling
Links to disambiguation pages	28	resolved automatically
Link to the wrong article (Long Beach)	1	fixed manually
Links to missing articles	3	rejected
Documents under 100 words	164	rejected
Team rows sharing one article	40	merged
Undocumented numeric `status`	–	inferred from `detail`
HTML entities in names	1	decoded

Table 2: Data quality issues and how they were handled

2.4 Final Dataset

The final collection comprises 2058 documents of four types and 59 distinct fields, stored in a 22.1 MB JSON file (Table 3). Each document combines structured fields from Jolpica, used for filters, with text fields from Wikipedia, which are the ones searched (Table 4). All documents also keep the article URL, the revision identifier and their word count. An abridged example of a race document is shown in Appendix A.

	Races	Drivers	Teams	Circuits	Total
Jolpica records	1162	818	168	77	2225
Accepted	1116	719	147	76	2058
Rejected	46	99	21	1	167
Fields	34	27	24	25	59
Median words	827	690	1082	1238	–

Table 3: Final collection. The total of fields counts distinct fields, since 9 are shared by all types

2.5 Conceptual Model

The conceptual model (Figure 3) describes the documents produced by the pipeline. An abstract **Document** holds the fields shared by all types (identifier, title, word count, Wikipedia identifiers and the `summary`). It is specialized into four classes, **RaceDocument**, **DriverDocument**, **TeamDocument** and **CircuitDocument**, each with its own structured attributes and text sections. Identifiers are canonical and readable, such as `race_2021_22` or `driver_hamilton`, and only documents with at least 100 words are accepted into the collection.

Each race has several drivers and at least two teams, and each driver participates in one or more races and drives for one or more

Type	Structured fields	Text fields
Race	season, date, circuit, country, winner, pole, podium, fastest lap, starters, finishers, retirements	summary, background, qualifying, race, post_race
Driver	nationality, birth date, seasons, teams, wins, podiums, poles, championships	summary, biography
Team	names, nationality, seasons, wins, podiums, championships	summary, history
Circuit	locality, country, coordinates, Grands Prix, most wins	summary, history, layout, events

Table 4: Main fields of each document type

teams. A team has at least one driver, not necessarily two: 47 Jolpica teams had a single driver in their whole history, mostly private entries of the 1950s. A team may have no linked race, which happens for three teams whose races were all rejected. Each race is held at exactly one circuit, which hosts one or more races; in the documents this link is optional, because the Pedralbes circuit was rejected and its two races have no circuit document. All relations are stored as lists of identifiers in both directions (e.g., `driver_ids` in a race and `race_ids` in a driver), allowing the search system to move from a document to the related ones.

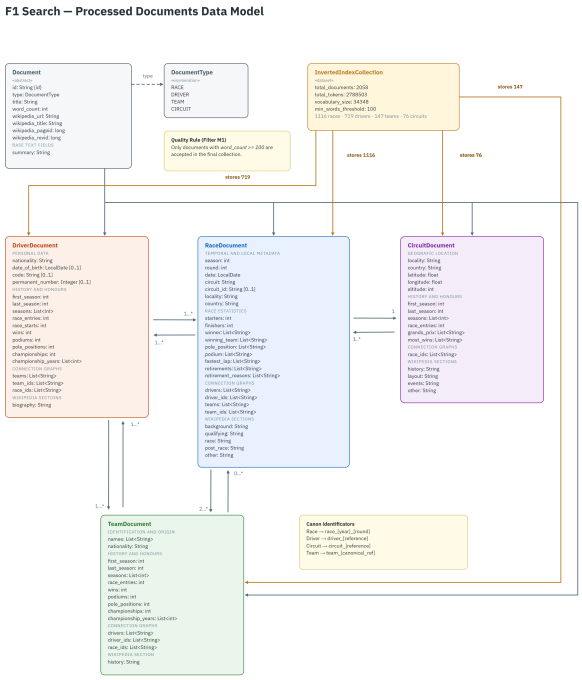


Figure 3: Domain conceptual model

2.6 Data Characterization

Document length is very uneven (Figure 4). In every type the mean is well above the median, pulled up by a few long articles: the

longest driver document (Kimi Räikkönen) has 14454 words, twenty times the median. For drivers, length grows with the career: the number of race entries and the number of words have a Spearman correlation of 0.69 (Figure 11). For races, length depends on the period (Figure 5): the median goes from about 330 words in the 1960s to 1849 in the 2000s, almost six times more. The 2000s are the peak, which suggests that these seasons were written about as they happened by a large community of editors, while older races were documented later and with fewer sources. This matters for the search system: a long document has more chances of matching a query, so a ranking that does not normalize length well will tend to favor recent races.

The collection has 2.79 million tokens and a vocabulary of 34348 terms, 37% of which appear only once. Stopwords make up 45% of all tokens. Term frequency follows Zipf's law [5] between ranks 10 and about 2000 (Figure 9), with the usual deviations at both ends: the most frequent words are less frequent than the law predicts, and the tail falls faster. The vocabulary grows sublinearly, as Heaps' law predicts (Figure 10). As documents are read grouped by type, the growth almost stops during the races, which reuse the names and terms of the biographies, and rises again with the teams.

After removing stopwords, "championship", "prix", "race" and "formula" appear in about 95% of the documents, so they will have little weight in a ranking based on inverse document frequency. The terms specific to each type are more telling (Figure 12): races use "lap", "pit", "qualifying" and "tyres", teams "engine", and circuits "track" and "turn".

To see how the language of the race reports changed over time, we extracted the keywords of each decade with TF-IDF [5], using scikit-learn and removing the names of drivers, teams and circuits, which would otherwise dominate. Table 5 shows that the keywords follow the history of the sport: privateer and works entries in the 1950s, turbo engines in the 1980s, refuelling stops and Friday sessions in the 2000s, and tyre compounds, penalties and the safety car in the 2010s. Some noise remains, such as "utc", from the start times given in recent articles.

Decade	Keywords
1950s	manuel, british, entered, privateer, argentine
1960s	manufacturers, cup, british, works, entered
1970s	cup, manufacturers, austrian, south, swedish
1980s	tag, turbo, brazilian, frenchman, italian
1990s	pre, report, failed, spun, failure
2000s	friday, constructors, stops, sessions, saturday
2010s	turn, penalty, tyre, compound, safety
2020s	utc, local, tyre, event, topped

Table 5: Top TF-IDF keywords of race documents by decade

In the structured fields, engine failures are the most common retirement reason (1889 cases), but accidents, collisions and spins together exceed 2500 (Figure 7). British (142) and American (131) drivers are the most common (Figure 8), the latter because the Indianapolis 500 counted for the championship until 1960. Finally, every race has a summary and 85% have a race report, but only 42% describe qualifying and 27% the aftermath (Figure 6), so some queries can only be answered for part of the collection, mostly recent seasons.

3 SEARCH SCENARIOS

Table 6 lists the information needs used to evaluate the system. They were chosen so that they depend on the text and not only on the structured fields, cover three document types, can be expressed with varied vocabulary, and have answers in the collection, which we confirmed by finding relevant documents for each. For each need we defined what makes a document relevant and what does not. In N2, for example, a race is relevant only if it was affected by rain **and** had an unusual result, such as the 1996 Monaco Grand Prix, where only three cars finished. A wet race won by the favorite is not relevant.

ID	Information need	Type
N1	Races decided by a pit-stop or tyre strategy	race
N2	Wet races with a surprising result	race
N3	Safety car periods that changed the result	race
N4	Collisions between teammates	race
N5	Race weekends with a fatal accident	race
N6	Disqualifications for technical infringements	race
N7	Drivers who returned after a serious injury	driver
N8	Drivers who succeeded at Le Mans or Indianapolis	driver
N9	Teams that collapsed for financial reasons	team
N10	Technical innovations that were later banned	team

Table 6: Information needs

Some needs are hard on purpose. In N1, the same idea appears as "undercut", "two-stop strategy" or "gambled on not needing a pit stop"; in N4, the text often gives only the two names, so the `teams` field is needed; N7 has few documents that use the words of the query, even though there are well-known cases such as Niki Lauda or Felipe Massa; and in N10, "banned" mostly returns the ban on turbo engines, which affected every team and is not relevant. These cases expose the limits of a search based only on words.

4 CONCLUSION AND FUTURE WORK

We built a reproducible collection of 2058 Formula One documents, characterized it and defined ten information needs.

The most challenging part was turning Wikipedia into usable text: the API blocks one-request-per-page clients, the wikitext mixes prose with tables and templates, and some links led to disambiguation pages. These were solved with batched and cached requests, a parser that keeps only the prose, and automatic link resolution.

As future work, the collection will be indexed in Apache Solr [9], comparing a single-field baseline with a setup that weights the most informative fields and filters by structured fields, both evaluated with precision at k and mean average precision. The harder information needs will then serve to compare this word-based search with semantic search.

References

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A Example Document

Abridged race document (`race_2021_22`); long lists and texts are cut with "…".

```
{
 "id": "race_2021_22",
 "type": "race",
 "title": "2021 Abu Dhabi Grand Prix",
 "season": 2021, "round": 22,
 "date": "2021-12-12",
 "circuit": "Yas Marina Circuit",
 "circuit_id": "circuit_yas_marina",
 "country": "UAE",
 "winner": ["Max Verstappen"],
 "winning_team": ["Red Bull"],
 "pole_position": ["Max Verstappen"],
 "podium": ["Max Verstappen",
   "Lewis Hamilton", "Carlos Sainz"],
 "starters": 19, "finishers": 14,
 "retirements": ["Sergio Perez: Engine",
   "Nicholas Latifi: Accident", ...],
 "driver_ids": ["driver_max_verstappen",
   "driver_hamilton", ...],
 "team_ids": ["team_red_bull",
   "team_mercedes", ...],
 "summary": "The 2021 Abu Dhabi Grand
   Prix was a Formula One motor race...",
 "race": "The race started at 17:00
   local time on Sunday...",
 "post_race": "In the laps before the
   safety car, Hamilton was leading...",
 "word_count": 5927,
 "wikipedia_revid": 1373349274
}
```

B Additional Figures

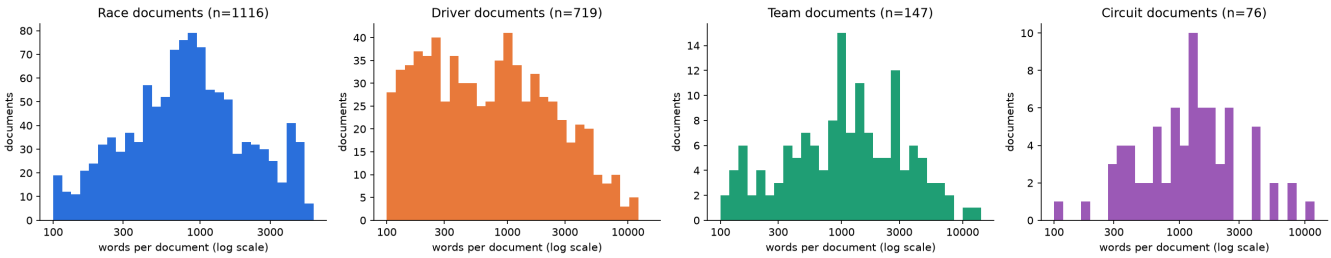


Figure 4: Words per document, by document type (log scale)

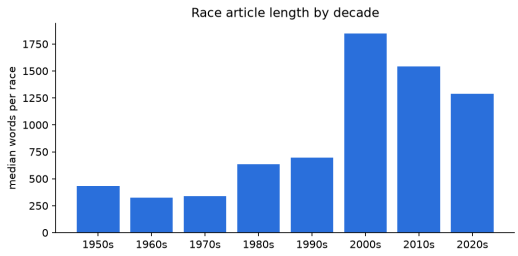


Figure 5: Median words per race document, by decade

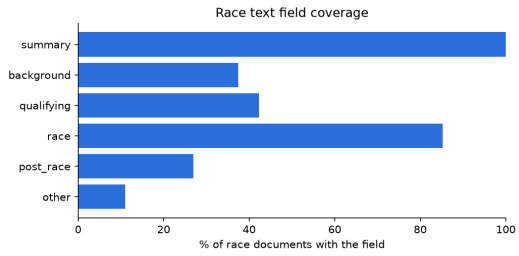


Figure 6: Share of race documents with each text field

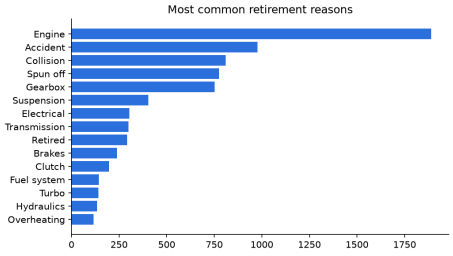


Figure 7: Most common retirement reasons

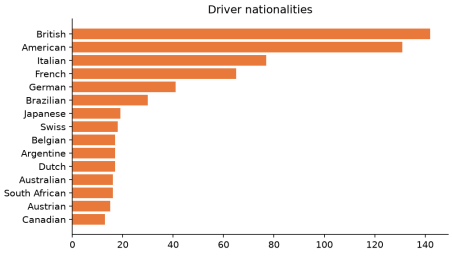


Figure 8: Driver nationalities

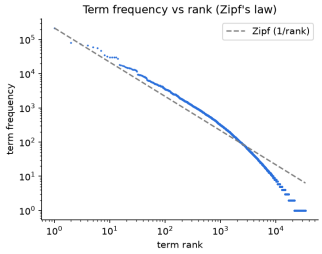


Figure 9: Term frequency versus rank

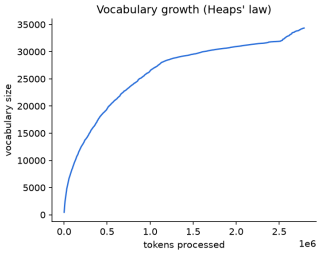


Figure 10: Vocabulary growth

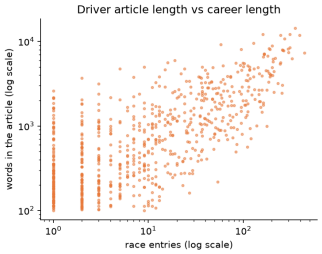


Figure 11: Driver article length versus race entries

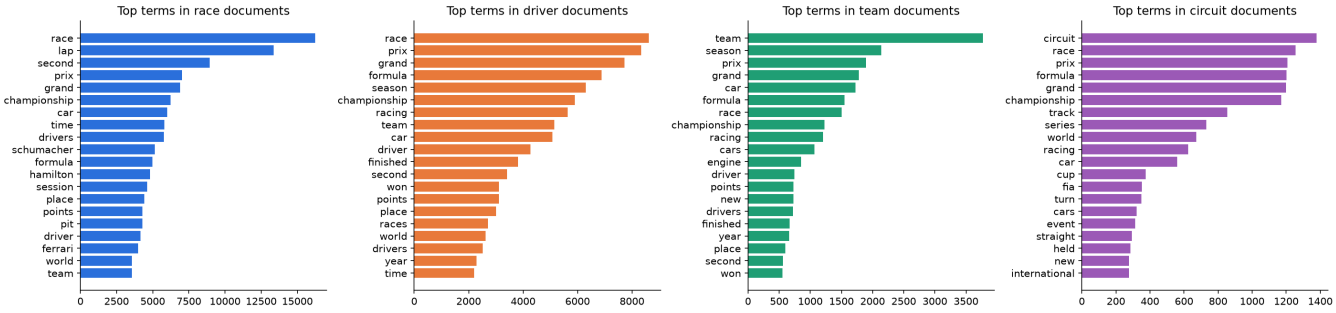


Figure 12: Most frequent terms per document type, without stopwords